

Advice for taking the COMS10015: Computer Architecture examination

For the 2025/26 academic year.

The ILOs can be found on the [Unit Catalogue](#). The coursework primarily tests the first Intended Learning Outcome (ILOs), which can be considered the easier of the ILOs. The examination therefore primarily tests the other three ILOs.

NOTE

The ILOs expect you to be able to: "recall and apply", "explain", and "explain trade-offs", and "demonstrate".

This means that you will be expected to be able to repeat **what** things are, but also explain the **why** behind their design. This requires that you build up both the technical knowledge of the concepts, but also develop an understanding of the advantages and disadvantages of the design of those concepts, and what limitations or goals were in place which motivated those concepts (for example, we introduce micro-architectural optimisations such as caches to optimise the performance of applications, at the expense of including a complex sub-system hidden from applications).

Much of the field of Computer Architecture is an Engineering discipline. It is highly practical, taking well defined concepts and technologies, and combines them together to build something useful. It creates and defines layers of abstraction so that it becomes easier to build more complex systems on top of lower-level systems (a natural example here is the Instruction Set Architecture, which defines the hardware/software interface).

TIP

Many students find the exam more challenging than the coursework - make sure you prepare appropriately.

Exam Weighting

The summative exam is worth 70% of the unit overall. This means your final mark is made up of a combination of your coursework and exam, 30% and 70% respectively (i.e., a weighted average of the two marks). There are no further restrictions on the assessments, as you may have had in other units (e.g., a "must-pass" assessment). However, it is highly recommended that you achieve your best mark in both assessments so that:

- you set a good foundation of knowledge in your first year (there is a correlation between your final degree award and your first year marks, even though they are not included in the final degree classification).
- you give yourself the best opportunity to pass the unit overall, as you need to at least do well enough in both to ensure the overall average mark exceeds the pass mark.

Exam Content

The unit is structured roughly in four quarters - two in TB-1, two in TB-2 - aligned to the four 6-week periods broken by Consolidation and Revision weeks. The coursework primarily tested the material taught in the first quarter (Weeks 1-6). The exam primarily tests the material taught in the other three quarters (Weeks 7-24). However, you may be tested on and require knowledge of the concepts from the entire course.

NOTE | We will not test your knowledge of Verilog in the examination.

Verilog is a description language for digital circuits. You used it to design digital circuits in your coursework. Therefore the examination will not require you to read or write any Verilog. The examination could require you to draw or interpret a digital circuit using other notation, e.g., logic gates.

Exam Style

The exam is a closed-book exam. Therefore you cannot bring additional materials to the exam. A standard non-programmable calculator is allowed.

The exam will consist of a number of Short Answer questions. The key thing to note is that the answer will therefore require you to write a relatively small amount. The exact amount that is appropriate is difficult to define, as it depends on the question, the number of marks available for that question, and what else the question might ask you to provide.

TIP | A possible rule of thumb is you should write roughly one sentence per mark available.

IMPORTANT | Writing more than is necessary won't lose you marks if everything you write is relevant to answering the question.

But you should be careful to stay on topic. Don't continue writing to show off your knowledge if it is not relevant to the question. You should also make sure you do not end up writing so much that you contradict yourself in the answer. Aim to write the answers clearly and concisely.

NOTE | Spelling, grammar, etc, does not matter: you do not lose marks for poor English.

What we are testing is your depth and range of knowledge. Therefore as long as you are clear enough to be understood, that's fine.

IMPORTANT | Write down the things the questions ask you to give. Read the question: clarify (e.g., note down) what is being asked of you and the marks available for it before starting.

For example, if the question asks for an example, give one. Here we can refer back to the previous note around length of answer: if you are asked for one example and give two, one of which is correct, one is not, then it is difficult to know whether you understand the concept or not.

Therefore, be confident in providing the number of examples asked for.

If the question asks for a justification, provide a reason or explanation. It is very important that you **do not** simply rephrase the question as your *only* explanation. Paraphrasing back the question does not demonstrate any understanding, and therefore an answer of this form is worth zero marks. Therefore a useful way to tackle these questions is to first provide definitions (i.e., "recall") of the concepts in the question. Then, use these definitions in the situation specified in the question.

TIP

Try to write something for each and every question, such as defining the terms used in the question.

A common exam question is to classify statements as true or false, and provide a justification for why that statement is true or false. One approach to solving this question is to provide "proof by counter example": if you can find an example which disproves the statement, then that will be sufficient. However, if you think a statement is true, then you need to show that it is true in all (relevant) cases - a single example is not enough. Think about how algebraic equations work: we use letters to show mathematical relationships hold for all values of that letter; you can take a similar approach here. For example, use letters like x to represent binary numbers. One way to start to tackle these questions is to try some examples, and try to spot a general pattern. This may be awarded partial marks in the exam, so show your working, but may not be awarded full marks in the cases where you need to prove things in the general case.

The use of mathematics or diagrammatic-style answers (such as drawing diagrams or figures), can help with your written answers. If the question asks you explicitly to draw a diagram, of course you should attempt one. But it may help you demonstrate clear understanding in other cases, although it is not mandatory or necessarily expected. Where used appropriately, they can convey a lot of information in a clear and concise manner.

Exam preparation

You should remember that "A 20-credit point unit is equivalent to 200 hours of notional student learning time comprised broadly of teaching time, independent study, assessment and feedback."

^[1] Independent study is a vital part of successful study at University. To help you get started with this, each lecture provides a list of suggested further reading - this can be found on the final slide.

They offer alternative ways to explain the same material from the lectures, which can be useful depending on how you learn. They also often go into further depth than is taught in the course. This can help you develop your understanding of the concepts, and see them applied in new and different circumstances (which is something you should expect to do in the examination).

Note that while the lecture materials contain everything you need to know for the assessments, it is recommended that you review the recommended additional materials as part of your exam preparation, along with others you may find on your own. The content of the additional material is not examinable, but will help reinforce the concepts from the lectures which are examinable.

The library has a wide array of books on Computer Architecture. One of the best, and most famous, is:

- Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy. This is available in Queen's Building Library at QA76.9.C643 PAT, and online at <https://bris.on.worldcat.org/oclc/1237754847>. You should also consult the online appendices.

There are multiple editions of this book as well as different versions which use a different Instruction Set Architecture for the examples. I haven't found a significant difference in these for the purposes of this course, so pick the one you can find, or the one with the ISA you recognise most.

Exam practice

There are **two question banks** available for you to practice. They are available in the top box on the Unit Webpage for each Teaching Block. Many of these practice questions have been taken from examinations in previous years. Whilst this is helpful to judge the difficulty and depth of knowledge needed in an examination, they are sometimes in a different style to the exam format this year. However, the process and steps needed to solve such multiple choice questions and solve a short answer question are much the same; only the notation of the answer is different.

This unit provided **four formative exams**. These offer questions, and solutions, on the exam topics. Remember however, these formative assessments were written to be taken over approximately one week, with full access to all your notes, the internet, etc, and therefore may not be indicative of the summative examination. The formative exams and solutions can be found in the relevant Announcements on BlackBoard.

There is also a past paper with solutions available on the [Unit Webpage](#). This is indicative of the exam format this year. The solution answers are representative of the mark scheme, and so are typically *longer* in length than what you might need to write in the exam. In particular, they often provide several example answers, which is important to ensure the full range of marks is available for your solution.

Information Resources

- The [unit organisation handbook](#), available on the [Unit Webpage](#) in the TB-1 material (first blue box on that page).
- Exam info lecture slides.
- Further advice about exam technique is available on the [Study Skills](#) website. I recommend the "exam papers and questions" module, in particular the "Exam questions - written answers" section of that.
- <https://www.bristol.ac.uk/students/your-studies/exams-assessments>

[1] <https://www.bristol.ac.uk/academic-quality/approve/approvalguidance/contacthours/>